

1. Derivation of Eq. (7)

We derive how the network outputs $(\mu_\epsilon, \ln \sigma_\epsilon)$ are mapped to the mean and standard deviation (μ_G, σ_G) of the Gaussian over $\mathbf{G}$ used in Eq. (7).

Consider one scalar dimension $G^{(d)}$ of the parameterized continuous graph representation $\mathbf{G}$, and let $\mu_t^{(d)}$ be the corresponding entry of the input mean at time t . From the Bayesian flow over parameters, the marginal distribution of $\mu_t^{(d)}$ conditioned on $G^{(d)}$ is

$$\mu_t^{(d)} \mid G^{(d)}, t \sim \mathcal{N}(\gamma(t) G^{(d)}, \gamma(t)(1 - \gamma(t))), \quad (1)$$

where $\gamma(t)$ is the function defined in the main text.

Therefore there exists $\epsilon^{(d)} \sim \mathcal{N}(0, 1)$ such that

$$\mu_t^{(d)} = \gamma(t) G^{(d)} + \sqrt{\gamma(t)(1 - \gamma(t))} \epsilon^{(d)}. \quad (2)$$

Solving for $G^{(d)}$ gives

$$G^{(d)} = \frac{\mu_t^{(d)}}{\gamma(t)} - \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \epsilon^{(d)}. \quad (3)$$

The core network at time t takes θ_t and optional conditioning as input and outputs

$$\epsilon = \Psi(\theta_t, \mathbf{c}, t), \quad (4)$$

which is split into two D -dimensional vectors $(\mu_\epsilon, \ln \sigma_\epsilon)$. For each dimension d we model the noise as

$$\begin{cases} \epsilon^{(d)} \mid \theta_t, t \sim \mathcal{N}(\mu_\epsilon^{(d)}, (\sigma_\epsilon^{(d)})^2), \\ \sigma_\epsilon^{(d)} = \exp(\ln \sigma_\epsilon^{(d)}), \end{cases} \quad (5)$$

Substituting this into Eq.(27), we obtain an affine transform of a Gaussian random variable:

$$G^{(d)} \mid \theta_t, t = \frac{\mu_t^{(d)}}{\gamma(t)} - \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \epsilon^{(d)}. \quad (6)$$

If $Z \sim \mathcal{N}(\mu_Z, \sigma_Z^2)$ and $a, b \in \mathbb{R}$, then $a + bZ \sim \mathcal{N}(a + b\mu_Z, b^2\sigma_Z^2)$. Applying this with

$$Z = \epsilon^{(d)}, \quad a = \frac{\mu_t^{(d)}}{\gamma(t)}, \quad b = -\sqrt{\frac{1 - \gamma(t)}{\gamma(t)}},$$

we obtain

$$\begin{cases} G^{(d)} \mid \theta_t, t \sim \mathcal{N}(\mu_G^{(d)}, (\sigma_G^{(d)})^2), \\ \mu_G^{(d)} = \frac{\mu_t^{(d)}}{\gamma(t)} - \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \mu_\epsilon^{(d)}, \\ (\sigma_G^{(d)})^2 = \frac{1 - \gamma(t)}{\gamma(t)} (\sigma_\epsilon^{(d)})^2, \end{cases} \quad (7)$$

and hence

$$\sigma_G^{(d)} = \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \sigma_\epsilon^{(d)} = \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \exp(\ln \sigma_\epsilon^{(d)}). \quad (8)$$

Stacking over all dimensions $d = 1, \dots, D$ and fixing $\mu_G = \mathbf{0}, \sigma_G = \mathbf{1}$ to avoid numerical issues when $\gamma(t)$ is extremely small, we yields:

$$\begin{aligned} \mu_G &= \begin{cases} \mathbf{0} & \text{if } t < t_{\min}, \\ \frac{\mu}{\gamma(t)} - \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \mu_\epsilon & \text{otherwise,} \end{cases} \\ \sigma_G &= \begin{cases} \mathbf{1} & \text{if } t < t_{\min}, \\ \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \exp(\ln \sigma_\epsilon) & \text{otherwise.} \end{cases} \end{aligned} \quad (9)$$

2. From KL flow loss to squared-error objective

Let $\mathbf{X} \in \mathbb{R}^D$ be the continuous parameterization of the atom features of a graph G (the bond case is analogous), and let $\mathbf{Y} \in \mathbb{R}^D$ denote the noisy message received from the sender. The continuous-time flow objective used by TopBF on $\mathbf{X}$ is

$$\mathcal{L}_{\text{flow}}(\mathbf{X}) = \int_0^1 \mathbb{E}_{\boldsymbol{\theta}_t \sim p_F(\boldsymbol{\theta}|\mathbf{X};t)} \left[D_{\text{KL}}(p_S(\mathbf{Y} | \mathbf{X}; \alpha(t)) \| p_R(\mathbf{Y} | \boldsymbol{\theta}_t; t, \alpha(t))) \right] dt. \quad (10)$$

By construction (Sec. 3.2), at each time t the sender and receiver for atoms are Gaussians with the same covariance:

$$p_S(\mathbf{Y} | \mathbf{X}; \alpha(t)) = \mathcal{N}(\mathbf{Y} | \mathbf{X}, \alpha(t)^{-1} \mathbf{I}), \quad (11)$$

$$p_R(\mathbf{Y} | \boldsymbol{\theta}_t; t, \alpha(t)) = \mathcal{N}(\mathbf{Y} | \hat{\mathbf{k}}_{\mathbf{X}}(\boldsymbol{\theta}_t, t), \alpha(t)^{-1} \mathbf{I}), \quad (12)$$

where $\hat{\mathbf{k}}_{\mathbf{X}}(\boldsymbol{\theta}_t, t)$ is the vector of expected bin centres of the output distribution p_O at time t .

For two Gaussians with identical covariance $\Sigma = \alpha^{-1} \mathbf{I}$,

$$\begin{aligned} D_{\text{KL}}(\mathcal{N}(\mathbf{m}_1, \Sigma) \| \mathcal{N}(\mathbf{m}_2, \Sigma)) \\ &= \frac{1}{2} (\mathbf{m}_1 - \mathbf{m}_2)^\top \Sigma^{-1} (\mathbf{m}_1 - \mathbf{m}_2) \\ &= \frac{\alpha}{2} \|\mathbf{m}_1 - \mathbf{m}_2\|_2^2. \end{aligned} \quad (13)$$

Setting $\mathbf{m}_1 = \mathbf{X}$ and $\mathbf{m}_2 = \hat{\mathbf{k}}_{\mathbf{X}}(\boldsymbol{\theta}_t, t)$ and substituting into Eq. (34) gives

$$\mathcal{L}_{\text{flow}}(\mathbf{X}) = \frac{1}{2} \int_0^1 \alpha(t) \mathbb{E}_{\boldsymbol{\theta}_t \sim p_F(\boldsymbol{\theta}|\mathbf{X};t)} [\|\mathbf{X} - \hat{\mathbf{k}}_{\mathbf{X}}(\boldsymbol{\theta}_t, t)\|_2^2] dt. \quad (14)$$

The channel accuracy is parameterized by a monotonically increasing function $\beta(t)$ with $\beta(0) = 0$ and $\beta(1) = \sigma_{1_{\mathbf{X}}}^{-2}$, and

$$\alpha(t) = \frac{d}{dt} \beta(t). \quad (15)$$

We use

$$\beta(t) = \sigma_{1_{\mathbf{X}}}^{-2t} \Rightarrow \alpha(t) = -2 \ln \sigma_{1_{\mathbf{X}}} \sigma_{1_{\mathbf{X}}}^{-2t}. \quad (16)$$

Thus

$$\frac{1}{2} \alpha(t) = -\ln \sigma_{1_{\mathbf{X}}} \sigma_{1_{\mathbf{X}}}^{-2t}, \quad (17)$$

and Eq. (38) becomes

$$\mathcal{L}_{\text{flow}}(\mathbf{X}) = -\ln \sigma_{1_{\mathbf{X}}} \int_0^1 \mathbb{E}_{\boldsymbol{\theta}_t \sim p_F(\boldsymbol{\theta}|\mathbf{X};t)} \left[\frac{\|\mathbf{X} - \hat{\mathbf{k}}_{\mathbf{X}}(\boldsymbol{\theta}_t, t)\|_2^2}{\sigma_{1_{\mathbf{X}}}^{2t}} \right] dt. \quad (18)$$

Writing the integral as an expectation over $t \sim \mathcal{U}(0, 1)$ yields the continuous-time loss

$$\mathcal{L}_{\infty}(\mathbf{X}) = -\ln \sigma_{1_{\mathbf{X}}} \mathbb{E}_{t \sim \mathcal{U}(0,1), \boldsymbol{\theta}_t \sim p_F(\boldsymbol{\theta}|\mathbf{X};t)} \left[\frac{\|\mathbf{X} - \hat{\mathbf{k}}_{\mathbf{X}}(\boldsymbol{\theta}_t, t)\|_2^2}{\sigma_{1_{\mathbf{X}}}^{2t}} \right]. \quad (19)$$

The same derivation for the continuous bond representation $\mathbf{A}$ with its own target scale $\sigma_{1_{\mathbf{A}}}$ gives $\mathcal{L}_{\infty}(\mathbf{A})$; the total training loss is

$$\mathcal{L}_{\mathbf{G}} = \mathcal{L}_{\mathbf{X}} + \mathcal{L}_{\mathbf{A}}.$$

3. Proof of Theorem 1

Proof Let $e = (i, j)$ and $e' = (u, v)$ be two distinct bonds and denote $D := C_{e,e'}^A = \min_{p \in \{i,j\}, q \in \{u,v\}} \text{dist}_G(p, q)$.

$$\text{dist}_{\mathcal{L}(G)}(e, e') \leq |(e, (p_0, p_1), \dots, (p_{D-1}, p_D), e')| \quad (20)$$

$$= D + 1 = 1 + C_{e,e'}^A, \quad (21)$$

where $(p_0, \dots, p_D)$ is a shortest G -path from an endpoint $p_0 \in e$ to an endpoint $p_D \in e'$. For any shortest $\mathcal{L}(G)$ -path $(e = e_0, \dots, e_L = e')$,

$$C_{e,e'}^A = \min_{p \in e, q \in e'} \text{dist}_G(p, q) \leq \text{dist}_G(p_0, p_L) \leq L - 1 = \text{dist}_{\mathcal{L}(G)}(e, e') - 1. \quad (22)$$

Combining yields $\text{dist}_{\mathcal{L}(G)}(e, e') = 1 + C_{e,e'}^A$.

For any coupling Π with total mass 1,

$$\langle \Pi, \text{dist}_{\mathcal{L}(G)} \rangle = \langle \Pi, C^A \rangle + 1, \quad (23)$$

so the additive shift does not change the optimal transport plan. $\square$

4. From Topology Matching to Class-Wise Transport Coupling

This appendix complements Sec.4.5 by making explicit the optimization principle behind Eqs. (16) and (17). Rather than penalizing misclassification point-wise, we seek the minimum transport cost required to reshape predicted categorical mass into the ground-truth mass along the geodesic geometry of the molecular graph.

Let P_X^t and P_A^t denote the node/bond categorical probabilities induced by θ_t through Eq. (11), and let Q_X, Q_A be the one-hot targets derived from the ground-truth discrete graph. We only couple classes that are present in the target to avoid ill-posed normalization:

$$\mathcal{K}_X^+(G) = \left\{ k \in [K_X] : \sum_{i=1}^N Q_X(i, k) > 0 \right\}, \quad (24)$$

$$\mathcal{K}_A^+(G) = \left\{ k \in [K_A] : \sum_{e \in \mathcal{E}} Q_A(e, k) > 0 \right\}, \quad (25)$$

and for bonds we exclude the no-bond class from $\mathcal{K}_A^+(G)$, as stated in Sec.4.5.

Let m_X and m_A be the valid masks. For each class k , we form class-wise mass fields on valid indices by simplex normalization:

$$\mathbf{a}_k^X = \text{Norm}(P_X^t(:, k) \odot m_X), \mathbf{b}_k^X = \text{Norm}(Q_X(:, k) \odot m_X), \quad (26)$$

$$\mathbf{a}_k^A = \text{Norm}(P_A^t(:, k) \odot m_A), \mathbf{b}_k^A = \text{Norm}(Q_A(:, k) \odot m_A), \quad (27)$$

where $\text{Norm}(v) = v / \langle \mathbf{1}, v \rangle$ whenever $\langle \mathbf{1}, v \rangle > 0$.

Given the geodesic ground costs C^X and C^A from Eq. (14) and (15), we follow our topology-matching principle that for each semantic class, transport the predicted mass to the target mass with minimum geodesic cost, then aggregate these costs across classes. This yields Eqs. (16) and (17).

4.1. A single optimization view.

The class-wise aggregation admits a clean equivalence to one OT problem on a product space, which clarifies what is being optimized and why it is transport-coupled.

Theorem 1 (Block-diagonal OT equivalence) Fix $C \in \mathbb{R}_+^{N \times N}$ and $\varepsilon > 0$. Let $\{\mathbf{a}_k\}_{k=1}^K, \{\mathbf{b}_k\}_{k=1}^K \subset \Delta^{N-1}$ be class-wise masses. Define measures on the product index set $[N] \times [K]$ by $\mu(i, k) = \frac{1}{K} \mathbf{a}_k(i)$ and $\nu(i, k) = \frac{1}{K} \mathbf{b}_k(i)$. Consider the cost on the product space:

$$\bar{C}((i, k), (j, k')) = \begin{cases} C_{ij}, & k = k', \\ +\infty, & k \neq k'. \end{cases} \quad (28)$$

Then the entropic OT objective on the product space decomposes as

$$\mathcal{W}_\varepsilon(\mu, \nu; \bar{C}) = \frac{1}{K} \sum_{k=1}^K \mathcal{W}_\varepsilon(\mathbf{a}_k, \mathbf{b}_k; C). \quad (29)$$

Proof We prove the decomposition by two chains of inequalities.

$$\mathcal{W}_\varepsilon(\mu, \nu; \bar{C}) = \min_{\Pi \geq 0} \langle \Pi, \bar{C} \rangle + \varepsilon \sum_{(i, k), (j, k')} \Pi_{(i, k), (j, k')} (\log \Pi_{(i, k), (j, k')} - 1) \quad (30)$$

$$\text{s.t.} \quad \sum_{j, k'} \Pi_{(i, k), (j, k')} = \mu(i, k), \quad \sum_{i, k} \Pi_{(i, k), (j, k')} = \nu(j, k').$$

$$\mathcal{W}_\varepsilon(\mu, \nu; \bar{C}) \geq \min_{\substack{\Pi \geq 0 \\ \Pi_{(i, k), (j, k')} = 0, \ k \neq k'}} \langle \Pi, \bar{C} \rangle + \varepsilon \sum_{(i, k), (j, k')} \Pi_{(i, k), (j, k')} (\log \Pi_{(i, k), (j, k')} - 1) \quad (31)$$

$$= \min_{\{\Pi_k\}_{k=1}^K} \frac{1}{K} \sum_{k=1}^K \left(\langle \Pi_k, C \rangle + \varepsilon \sum_{i, j} (\Pi_k)_{ij} (\log (\Pi_k)_{ij} - 1) \right) - \varepsilon \log K \quad (32)$$

$$\text{s.t.} \quad \Pi_k \geq 0, \Pi_k \mathbf{1} = \mathbf{a}_k, \Pi_k^\top \mathbf{1} = \mathbf{b}_k, \forall k,$$

$$= \frac{1}{K} \sum_{k=1}^K \mathcal{W}_\varepsilon(\mathbf{a}_k, \mathbf{b}_k; C) - \varepsilon \log K. \quad (33)$$

$$\mathcal{W}_\varepsilon(\mu, \nu; \bar{C}) \leq \langle \Pi^*, \bar{C} \rangle + \varepsilon \sum_{(i, k), (j, k')} \Pi_{(i, k), (j, k')}^* (\log \Pi_{(i, k), (j, k')}^* - 1) \quad (34)$$

$$= \frac{1}{K} \sum_{k=1}^K \mathcal{W}_\varepsilon(\mathbf{a}_k, \mathbf{b}_k; C) - \varepsilon \log K, \quad (35)$$

where Π^* is constructed from the class-wise optimizers $\{\Pi_k^*\}$ by

$$\Pi_{(i,k),(j,k')}^* = \frac{1}{K} (\Pi_k^*)_{ij} \mathbf{1}[k = k'].$$

Combining Eqs. (57) – (59) yields

$$\mathcal{W}_\varepsilon(\mu, \nu; \bar{C}) = \frac{1}{K} \sum_{k=1}^K \mathcal{W}_\varepsilon(\mathbf{a}_k, \mathbf{b}_k; C) - \varepsilon \log K. \quad (36)$$

Since $-\varepsilon \log K$ is a constant independent of $\{\Pi_k\}$, it can be absorbed into the objective without affecting the optimal coupling. Therefore, up to an additive constant,

$$\mathcal{W}_\varepsilon(\mu, \nu; \bar{C}) \equiv \frac{1}{K} \sum_{k=1}^K \mathcal{W}_\varepsilon(\mathbf{a}_k, \mathbf{b}_k; C), \quad (37)$$

which is the stated decomposition. $\square$

4.2. Proof of Theorem 2

Proof Fix a class k and consider unregularized OT

$$\mathcal{W}_0(\mathbf{a}_k, \mathbf{b}_k; C) = \min_{\Pi \geq 0} \langle \Pi, C \rangle \quad \text{s.t.} \quad \Pi \mathbf{1} = \mathbf{a}_k, \quad \Pi^\top \mathbf{1} = \mathbf{b}_k. \quad (38)$$

Let $r = \sum_{i=1}^N Q(i, k)$ be the number of valid indices of class k in the target, with $I = \{i_1, \dots, i_r\}$ (predicted locations) and $J = \{j_1, \dots, j_r\}$ (target locations). In the hard-label limit,

$$\mathbf{a}_k = \frac{1}{r} \sum_{\ell=1}^r \mathbf{e}_{i_\ell}, \quad \mathbf{b}_k = \frac{1}{r} \sum_{\ell=1}^r \mathbf{e}_{j_\ell}. \quad (39)$$

The marginal constraints imply $\Pi_{uv} = 0$ unless $(u, v) \in I \times J$; define the restricted matrix $\tilde{\Pi} \in \mathbb{R}_+^{r \times r}$ and cost $\tilde{C} \in \mathbb{R}_+^{r \times r}$ by

$$\tilde{\Pi}_{\ell m} = r \Pi_{i_\ell j_m}, \quad \tilde{C}_{\ell m} = C_{i_\ell j_m}. \quad (40)$$

Then $\tilde{\Pi} \mathbf{1} = \mathbf{1}$ and $\tilde{\Pi}^\top \mathbf{1} = \mathbf{1}$, and

$$\langle \Pi, C \rangle = \sum_{\ell, m} \Pi_{i_\ell j_m} C_{i_\ell j_m} \quad (41)$$

$$= \frac{1}{r} \sum_{\ell, m} \tilde{\Pi}_{\ell m} \tilde{C}_{\ell m} \quad (42)$$

$$= \frac{1}{r} \langle \tilde{\Pi}, \tilde{C} \rangle. \quad (43)$$

Therefore

$$\mathcal{W}_0(\mathbf{a}_k, \mathbf{b}_k; C) = \frac{1}{r} \min_{\tilde{\Pi} \in \mathcal{B}_r} \langle \tilde{\Pi}, \tilde{C} \rangle, \quad \mathcal{B}_r = \{\tilde{\Pi} \geq 0 : \tilde{\Pi} \mathbf{1} = \mathbf{1}, \tilde{\Pi}^\top \mathbf{1} = \mathbf{1}\}. \quad (44)$$

Since the objective is linear, an optimum is attained at an extreme point of $\mathcal{B}_r$. By Birkhoff–von Neumann (Von Neumann 1953), the extreme points are permutation matrices P_π ($\pi \in \mathfrak{S}_r$), hence

$$\mathcal{W}_0(\mathbf{a}_k, \mathbf{b}_k; C) = \frac{1}{r} \min_{\pi \in \mathfrak{S}_r} \sum_{\ell=1}^r C_{i_\ell, j_{\pi(\ell)}}. \quad (45)$$

Thus, in the hard-label regime, class-wise OT reduces to within-class minimum-cost matching under the geodesic cost C , making long-range permutations strictly more expensive than local corrections. $\square$

Table 1 Hyper-parameters of TopBF.

Category	Hyper-parameter	Description	Value
Bayesian Flow	$\sigma_{1,\mathbf{X}}$	The standard deviation of $\mathbf{X}$ of the noise distribution	0.2
	$\sigma_{1,\mathbf{A}}$	The standard deviation of $\mathbf{A}$ of the noise distribution	0.2
	T	Sampling steps	200
	t_{min}	Minimum time	10^{-4}
QW Transport	λ_{QW}	The coefficient weighting the QW structural regularizer	0.1
	ϵ_{QW}	The entropic regularization smoothing the optimal transport plan calculation	0.2
	n_{QW}	The number of Sinkhorn iterations of the entropic Optimal Transport problem	30

5. Experimental Details

For our TopBF, we train our model with a constant learning rate 10^{-4} with AdamW optimizer and weight decay 10^{-12} , applying the exponential moving average (EMA) to the parameters. The core network $\Psi(\cdot)$ of TopBF is the graph transformer (Dwivedi and Bresson 2020; Vignac et al. 2023), with the inner parameters consistent across all baselines using it. Other critical hyper-parameters are presented in Table 1.

6. Supplementary Experiments

6.1. Computational Efficiency and Wall-Clock Time Analysis

We conduct a rigorous wall-clock time comparison against state-of-the-art baselines. We repeat the measurement and report the average.

Experimental Setup

To ensure a fair comparison, all runtime measurements were conducted under a strictly controlled hardware and software environment.

- **Hardware:** All models were trained and evaluated on a single NVIDIA A6000 GPU with an AMD EPYC 7763 (64 cores) @ 2.450GHz CPU.
- **Software:** Experiments were implemented in PyTorch 2.0.1 with CUDA 11.7.
- **Controls:** We fixed the batch size to 1,024 for QM9 and 256 for ZINC250. For sampling efficiency, we measured the total time required to generate 10,000 valid molecules with their own reported sampling steps.

We compare TopBF against three representative baselines: GDSS (Jo et al. 2022), DiGress (Vignac et al. 2023), and SID (Boget 2025). These methods represent the current standard for score-based and discrete generative models, respectively.

Table 2 Comparison of total sampling time. Lower is better.

Method	QM9 (min)↓	ZINC250k (min)↓
GDSS	1.42	26.47
DiGress	10.08	212.95
SID	1.58	25.78
TopBF (Ours)	1.33	23.32

As shown in Table 2, TopBF maintains its lead by generating 10,000 valid molecules in just 1.33 minutes for QM9 and 23.32 minutes for ZINC250k. These results indicate that TopBF achieves competitive or improved sampling throughput under our controlled setup. Since QW is used only as a training-time regularizer, it does not introduce additional cost during sampling.

6.2. Interpretability Analysis

QW regularization is intended to encourage topology-consistent (geodesically local) corrections. To make this effect observable beyond final metrics, we report controlled probing at fixed noise levels t . We evaluate four representative noise levels $t \in \{0.05, 0.2, 0.5, 0.8\}$.

For probing, we quantify locality via Sinkhorn-OT diagnostics under shortest-path cost: transport cost $\text{cost_avg}\downarrow$ and local-mass ratios within 1/2 hops $\text{localmass}_{1/2}\uparrow$, together with a correctness proxy $\text{avg_pX_gt}\uparrow$. Improve is relative improvement: for $\downarrow$, $(\text{Base} - \text{QW})/\text{Base}$; for $\uparrow$, $(\text{QW} - \text{Base})/\text{Base}$.

Across noise levels, probing verifies the intended mechanism: QW promotes geodesically local corrections (lower cost_avg , higher $\text{localmass}_{1/2}$).

The coupling matrix P transports the predicted node-type mass to the ground-truth mass with graph geodesic distance as the unit cost. Mass concentrated near small distances indicates that correcting node-type distributions primarily involves local adjustments on the graph, whereas diffuse off-diagonal mass implies long-range relocation. As shown in Figures 1 and 2, QW produces a noticeably more localized transport plan than the Base model at the same noise level, consistent with the reduced cost_avg and increased local-mass ratios in Table 3.

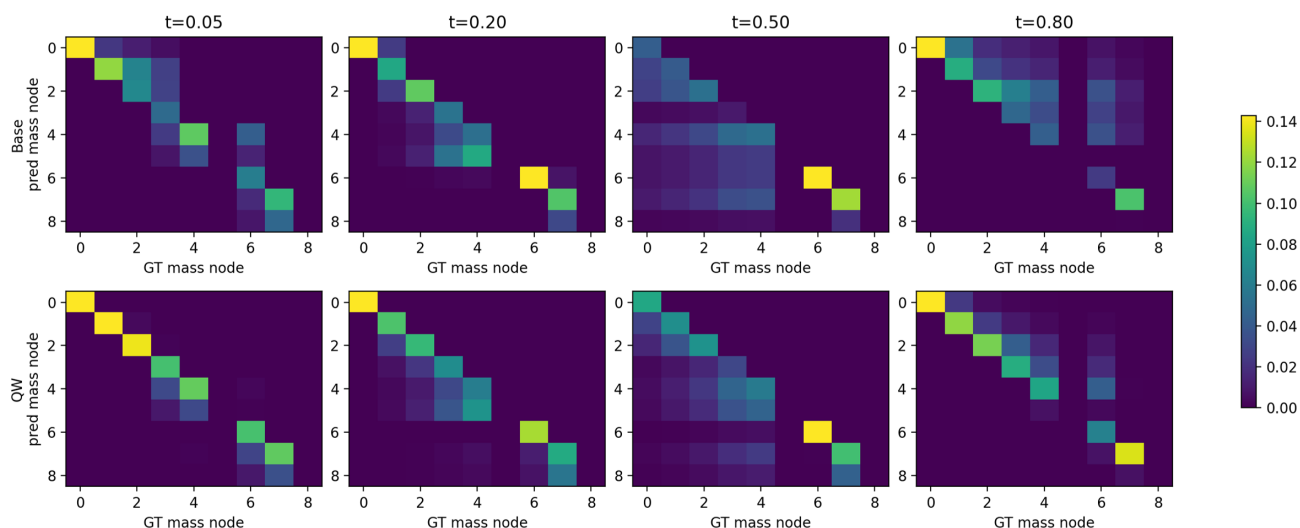


Figure 1 Sinkhorn coupling P between the predicted node-type marginals and the ground truth under the shortest-path (geodesic) cost, probed at $t \in \{0.05, 0.2, 0.5, 0.8\}$ on QM9 dataset.

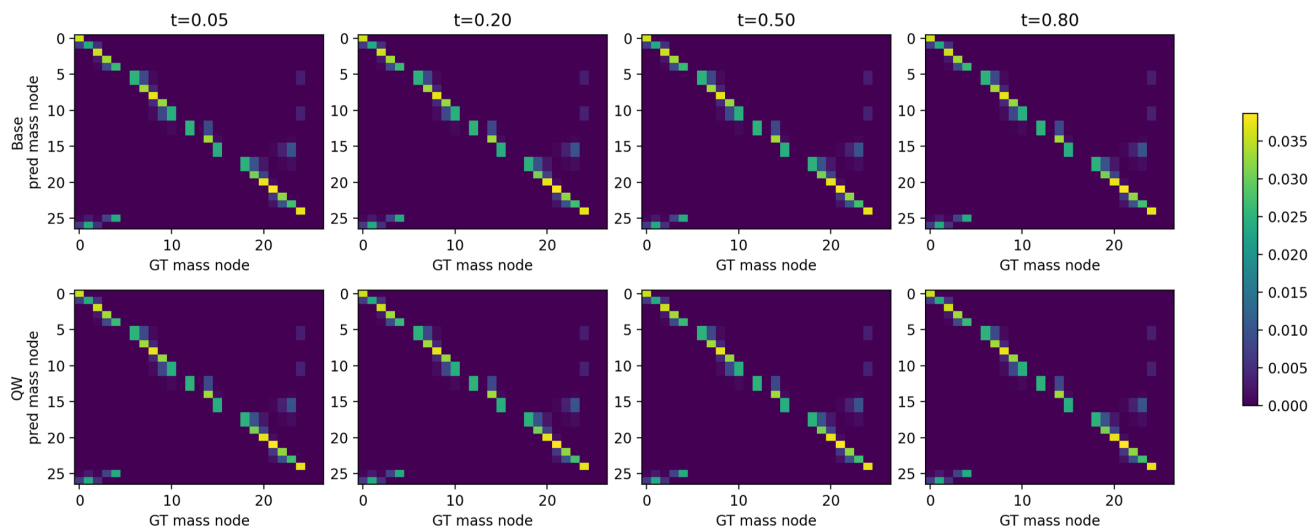


Figure 2 Sinkhorn coupling P between the predicted node-type marginals and the ground truth under the shortest-path (geodesic) cost, probed at $t \in \{0.05, 0.2, 0.5, 0.8\}$ on ZINC250k dataset.

7. Visualization of Generated Molecular Graphs

We demonstrate some randomly selected molecular graphs generated by TopBF on both QM9 and ZINC250k datasets in Figures. 3 and 4.

8. Pseudocode for TopBF

Pseudocode for essential functions in TopBF and continuous-time loss $\mathcal{L}_\infty$ are presented in Algorithms 1 and 2, while sampling procedure is provided in Algorithm 3, wherein NEAREST_CENTER compares the inputs to the nearest centers.

Table 3 Probing statistics under shortest-path cost at $t \in \{0.05, 0.2, 0.5, 0.8\}$ (Base vs. QW). QW reduces transport cost and increases local transport mass, indicating more geodesically local corrections.

t	cost_avg↓			localmass1↑			localmass2↑			avg.pX_gt↑		
	Base	QW	Improve	Base	QW	Improve	Base	QW	Improve	Base	QW	Improve
0.05	0.834	0.605	27.5%	0.810	0.852	5.2%	0.911	0.923	1.3%	0.328	0.495	50.9%
0.20	0.815	0.767	5.9%	0.800	0.813	1.6%	0.901	0.911	1.1%	0.347	0.602	73.5%
0.50	1.454	1.149	21.0%	0.612	0.698	14.1%	0.755	0.826	9.4%	0.422	0.581	37.7%
0.80	0.800	0.344	57.0%	0.761	0.913	20.0%	0.913	0.987	8.1%	0.552	0.707	28.1%

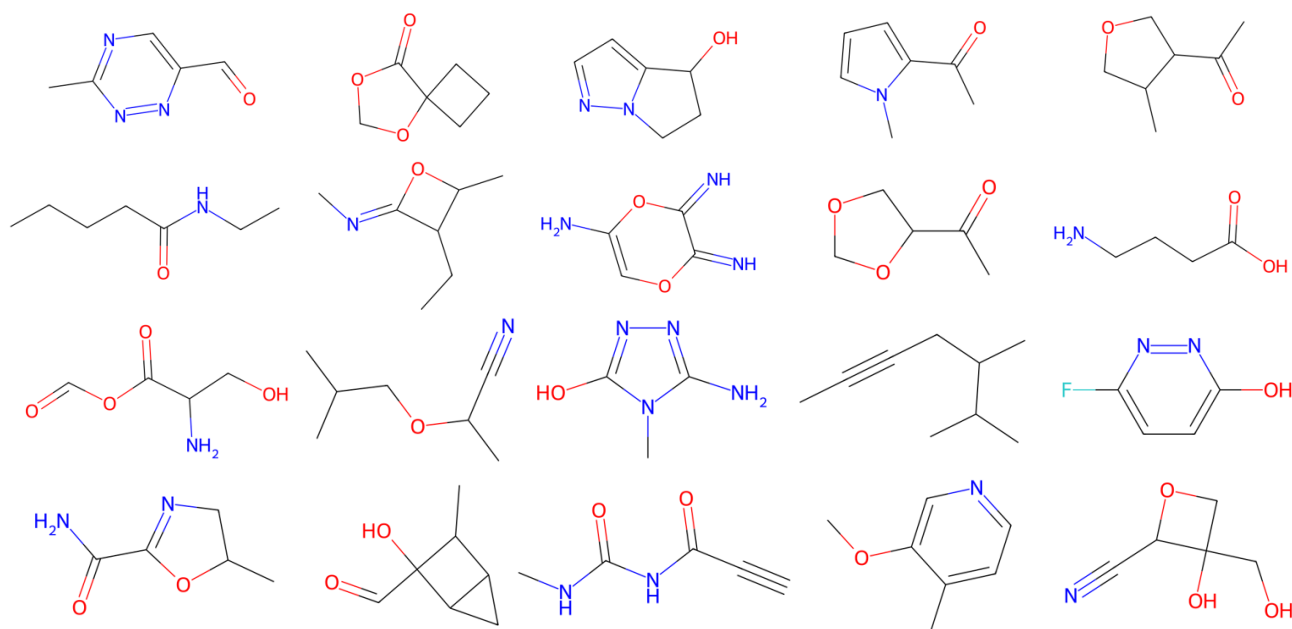


Figure 3 Molecular graphs generated by TopBF on QM9 dataset.

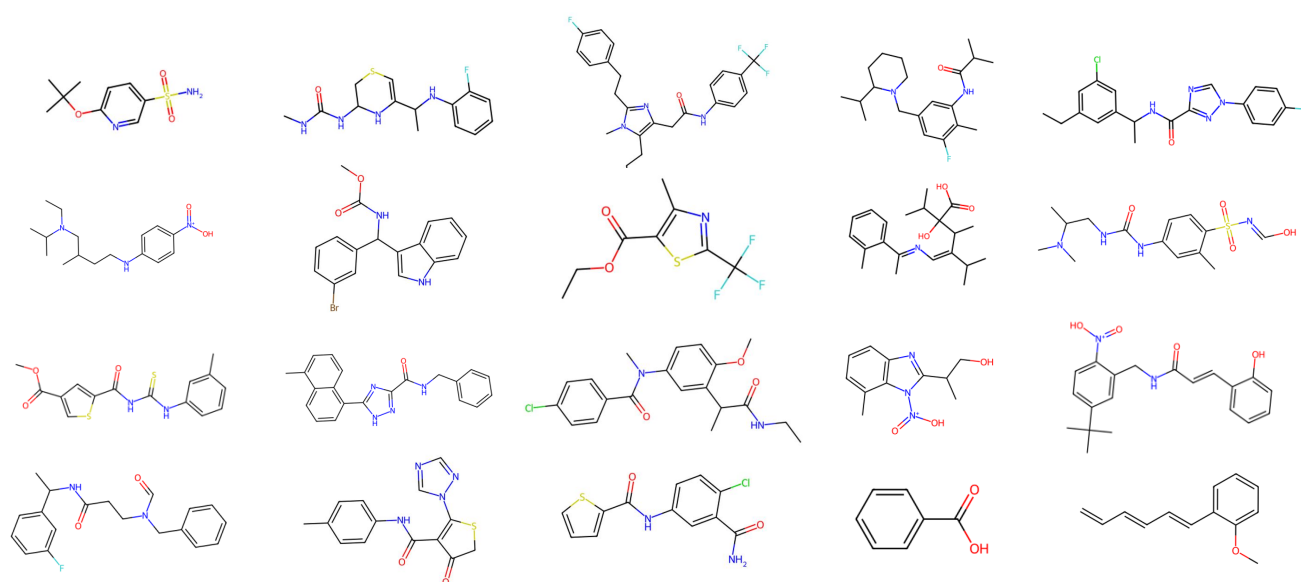


Figure 4 Molecular graphs generated by TopBF on ZINC250k dataset.

Algorithm 1 Functions for TopBF

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function CDF( $\mu, \sigma, x$ )
 $F(x) \leftarrow \frac{1}{2} \left[ 1 + \operatorname{erf}\left(\frac{x - \mu}{\sqrt{2}\sigma}\right) \right]$ 
if  $x \leq -1$  then
 $H(x) \leftarrow 0$ 
else if  $x \geq 1$  then
 $H(x) \leftarrow 1$ 
else
 $H(x) \leftarrow F(x)$ 
end if
return  $H(x)$ 
end function

function PRED( $\mu \in \mathbb{R}^{D \times K}, t \in [0, 1], K \in \mathbb{N}, \gamma(t) \in \mathbb{R}^+, t_{\min} \in \mathbb{R}^+$ )
#  $t_{\min}$  is set to 0.0001 by default
if  $t < t_{\min}$  then
 $\hat{\mu} \leftarrow 0, \hat{\sigma} \leftarrow 1$ 
else
 $\mu_\epsilon, \ln \sigma_\epsilon \leftarrow \Psi(\theta_t, \mathbf{c}, t)$ 
 $\hat{\mu} \leftarrow \frac{\mu}{\gamma(t)} - \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \mu_\epsilon$ 
 $\hat{\sigma} \leftarrow \sqrt{\frac{1 - \gamma(t)}{\gamma(t)}} \exp(\ln \sigma_\epsilon)$ 
end if
for  $d = 1, \dots, D$  do
for each  $k \in K$  do
 $p_O^{(d)}(k | \theta) \leftarrow \text{CDF}(\hat{\mu}^{(d)}, \hat{\sigma}^{(d)}, k_r) - \text{CDF}(\hat{\mu}^{(d)}, \hat{\sigma}^{(d)}, k_l)$ 
end for
end for
return  $p_O(\cdot | \theta_t)$ 
end function

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Algorithm 2 Training Algorithm

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Require:  $\sigma_1 \in \mathbb{R}^+, K \in \mathbb{N}$ 
Input: parameterized continuous graph  $\mathbf{G}$ 
 $t \sim \mathcal{U}(0, 1)$ 
 $\gamma \leftarrow 1 - \sigma_1^{2t}$ 
 $\mu \sim \mathcal{N}(\gamma(t), \gamma(t)(1 - \gamma(t))\mathbf{I})$ 
 $p_O(\cdot | \theta_t) \leftarrow \text{PRED}(\mu, t, K, \gamma(t))$ 
 $\mathbf{k}_c(\theta_t, t) \leftarrow \text{NEAREST\_CENTER}(p_O(\cdot | \theta_t))$ 
 $\mathcal{L}_\infty(\mathbf{G}) \leftarrow -\ln \sigma_1 \sigma_1^{-2t} \|\mathbf{G} - \mathbf{k}_c(\theta_t, t)\|_2^2$  return  $\mathcal{L}_\infty(\mathbf{G})$ 

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Algorithm 3 Sampling Algorithm

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Require:  $\sigma_1 \in \mathbb{R}^+$ , number of steps  $n, K \in \mathbb{N}$ , optional  $\mathbf{z}^*, \lambda \geq 0$ 
 $\mu_0 \leftarrow \mathbf{0}, \rho_0 \leftarrow 1$ 
for  $i = 1$  to  $n$  do
 $t \leftarrow \frac{i-1}{n}$ 
 $\gamma(t) \leftarrow 1 - \sigma_1^{2t}$ 
 $\theta_t \leftarrow (\mu_{i-1}, \rho_{i-1})$ 
if  $\mathbf{z}^*$  is not None and  $\lambda > 0$  then
 $\hat{\mathbf{z}}_t \leftarrow R_\varphi(\theta_t, t)$ 
 $\mathcal{L}_{\text{prop}} \leftarrow \|\hat{\mathbf{z}}_t - \mathbf{z}^*\|_2^2$ 
 $\mu_{i-1} \leftarrow \mu_{i-1} - \lambda \nabla_{\mu_{i-1}} \mathcal{L}_{\text{prop}}$ 
 $\theta_t \leftarrow (\mu_{i-1}, \rho_{i-1})$ 
end if
 $p_O(\cdot | \theta_t) \leftarrow \text{PRED}(\mu_{i-1}, t, K, \gamma(t))$ 
 $\alpha(t) \leftarrow \sigma_1^{-2i/n} (1 - \sigma_1^{2/n})$ 
 $\mathbf{k}_c(\theta_t, t) \leftarrow \text{NEAREST\_CENTER}(p_O(\cdot | \theta_t))$ 
 $\mathbf{Y} \sim \mathcal{N}(\mathbf{k}_c(\theta_t, t), \alpha(t)^{-1} \mathbf{I})$ 
 $\mu_i \leftarrow \frac{\rho_{i-1} \mu_{i-1} + \alpha(t) \mathbf{Y}}{\rho_{i-1} + \alpha(t)}$ 
 $\rho_i \leftarrow \rho_{i-1} + \alpha(t)$ 
end for
 $p_O(\cdot | \theta_1) \leftarrow \text{PRED}(\mu_n, 1, K, 1 - \sigma_1^2)$ 
 $\mathbf{k}'_c(\theta_1, 1) \leftarrow \text{NEAREST\_CENTER}(p_O(\cdot | \theta_1))$  return  $\mathbf{k}'_c(\theta_1, 1)$ 

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